

Quantifying Vocal Biomarkers in ALS and non-ALS Individuals

ESE 4970 Capstone Design Project Proposal Submitted to the Department of Electrical and Systems
Engineering

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I. Introduction

Problem Statement:

The objective of this project was to design a MATLAB algorithm that quantifies fundamental vocal features, provides informative indicators of vocal health, and is feasible to implement in a clinical setting for data collection. Specifically, the system will analyze the vocal characteristics of jitter, shimmer, harmonic-to-noise ratio (HNR), and per second harmonic contribution for ALS and non-ALS individuals using prolonged vowels \a\ and \i\. Baseline feature values will be established for non-ALS speakers across age and sex demographics and used as reference points for characterizing differences observed in ALS prolonged-vowel recordings.

Aims and Objectives:

Objective 1 – Quantifying Vocal Characteristics: Develop an open-source MATLAB algorithm that takes a sustained vowel recording as the input and outputs quantitative metrics of jitter, shimmer, HNR, and per second harmonic contribution. A hardware extension of the MATLAB algorithm should enable users to record a prolonged vowel in real time and compute all four metrics reliably using a microphone and computer-based acquisition system.

Objective 2 – Quantifying ALS-like Physiological Degeneration: Extend the algorithm to analyze quantifiable population differences between for ALS and non-ALS using the four vocal metrics. From this data, create a composite metric that attempts to classify whether an individual is in the ALS or non-ALS group using audio files from the Minsk2020 ALS database [1].

Key Outcomes:

Real-time analysis: When finished, the MATLAB algorithm was able to input an individual's live audio recording and output reliable values for all four quantitative metrics. The reliability of the system was quantified by observing plots that proved the correct information was being extracting from the individual's waveform and verifying that the quantitative metrics fell within the range of data seen in the Minsk2020 ALS dataset.

Increased variability in ALS individuals: Across jitter, shimmer, and per second harmonic contribution, the ALS group shows consistently higher variability. The Poincaré perpendicular SD was jitter, shimmer, and per second harmonic contribution changed from 0.0475 (non-ALS) to 0.0734 (ALS), 0.0924 to 0.1339, and 0.0048 to 0.0093, respectively. Additionally, ALS individuals have a lower HNR (group: 13.44 dB; men: 12.13 dB; females: 15.91 dB) than non-ALS individuals (grouped: 16.33 dB; men: 14.13dB; female: 17.87dB).

Success as an ALS vs non-ALS classifier: Combining all four metrics as a composite metric for ALS vs non-ALS classification and performing ROC analysis, the composite metric had an Area Under the Curve (AUC) of 0.76. This means the algorithm performs moderately well as a means of using prolonged \a\ and \i\ recording at least three seconds long to distinguish between individuals diagnosed with ALS and those without.

II. Background Justification

Bridge2AI Voice as a Biomarker of Health is a National Institutes of Health (NIH) funded initiative dedicated to the creation of standardized, high-quality voice datasets that support the development of artificial intelligence models for health assessment [2]. Rooted in the rapidly evolving field of vocal biomarker research, this initiative investigates vocal characteristics associated with neurological and neurodegenerative disorders, mood and psychiatric conditions, respiratory diseases, and voice disorders. Vocal biomarkers are particularly promising for clinical use due to their noninvasive nature, low cost, and broad applicability, making them viable for routine clinical use, including annual medical checkups.

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease for which vocal biomarkers show strong potential. The progression of ALS affects upper and lower motor neurons, often leading to dysarthria, defined as a motor speech disorder caused by impaired coordination of the bulbar muscles used in speech production [3, 4]. The paper “Classification of ALS patients based on acoustic analysis of sustained vowel phonations” demonstrates the ability to distinguish ALS speakers from healthy controls using signal processing techniques applied to prolonged-vowel recordings [5]. In that study, perturbation-based acoustic features were extracted from the open-source Minsk2020 ALS dataset to classify ALS speakers versus healthy controls [5].

However, current literature lacks open-source, transparency code and often uses overly aggressive classification strategies that may obscure whether observed differences reflect true disease-related vocal changes or demographic imbalance within the dataset. The finalized version of this capstone project produces an open-source MATLAB package that allows other engineers and researchers to confirm the validity of the metric algorithm and utilize the tool when helpful. Additionally, demographic controlled population statistics are visualized to better understand differences between audio produced by ALS and non-ALS individuals, removing the need to use overly complex classifications algorithms.

III. Methods

To differentiate between ALS and non-ALS individuals, vocal analysis of prolonged vowel phonations provides insight into neuromuscular changes that affect speech production. For individuals with ALS, sustaining a consistent pitch and loudness while holding a vowel can become more challenging due to reduced muscular control.

The following methods apply signal processing techniques to quantify irregularities in phonation and target areas of speech that give insight into a lack of neuromuscular control. The data collection task consists of sustained vowels /a/ and /i/, which produce approximately periodic signals that enable measurement of perturbations and irregularities in vocal function. Data is primarily sourced from the Minsk2020 ALS dataset, comprising 128 sustained-vowel phonations from 31 ALS and 33 non-ALS speakers [5]. Sex, age, and ALS status are available along with the raw audio files.

Quantifiable Vocal Characteristics Definitions

Jitter is defined as rapid, cycle-to-cycle variability in the timing of vocal fold vibrations during phonation and is commonly observed in ALS due to loss of muscular control. To calculate jitter, it is necessary to measure variation in consecutive period lengths and can be averaged over any number of periods [6]. For this analysis, local jitter, as shown in Equation (1), is used. Local jitter sums the absolute differences

between adjacent periods, T_i and T_{i+1} , and normalizes by the mean period, $\frac{1}{N} \sum_{i=1}^N T_i$, where N is the number of periods.

$$Jitter = \frac{1}{N-1} \cdot \frac{\sum_{i=1}^{N-1} |T_{i+1} - T_i|}{\frac{1}{N} \sum_{i=1}^N T_i} \quad (1)$$

Shimmer is defined as cycle-to-cycle variability in the maximum absolute amplitude of vocal fold vibrations during phonation [6]. Similar to jitter, reduced muscular control in the vocal tract can increase amplitude variability, especially during prolonged phonation. Equation (2) represents local shimmer and quantifies the average absolute difference in amplitude between adjacent periods, A_i and A_{i+1} , and then normalizes by the mean amplitude, $\frac{1}{N} \sum_{i=1}^N A_i$, across N periods [5].

$$Shimmer = \frac{1}{N-1} \cdot \frac{\sum_{i=1}^{N-1} |A_{i+1} - A_i|}{\frac{1}{N} \sum_{i=1}^N A_i} \quad (2)$$

Harmonic-to-Noise ratio (HNR) describes the ratio of harmonic energy to noise energy in a voice signal and is given by Equation (3) [7]. Harmonic and noise energy is normalized by the measuring bandwidth to ensure an accurate metric. In practice, HNR provides insight into how periodic versus noisy a sustained phonation is.

$$HNR(dB) = 10 \log_{10} \frac{\left(\frac{H_p}{BW_h}\right)}{\left(\frac{N_p}{BW_n}\right)} \quad (3)$$

Per Second Harmonic Power is a metric that quantifies the change in the harmonic distribution over one second audio intervals showed by Equation (4), where $d_{s,k} = |PC_s(k) - PC_{s-1}(k)|$ is the absolute difference of the power contributions of each harmonic frequency k compared to the power contributions of the corresponding frequency from the previous second, and N_d and N_k are the count of the differences and harmonic frequencies, respectively. Finally, $PC(k)$ is the power contributions of each harmonic frequency k .

$$Per\ Second\ Harmonic\ Power = \frac{\frac{1}{N_d} \sum d_s(k)}{\frac{1}{N_k} \sum PC_s(k)} \quad (4)$$

Block Diagram of Hardware System

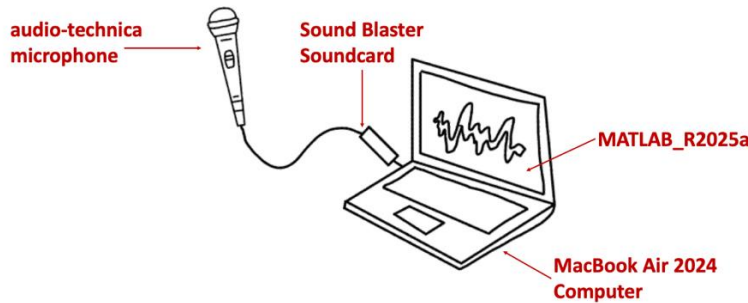


Figure 1: Hardware and software specifications and set-up for algorithm use.

Overview of Algorithm

Time Domain Analysis

1. An audio file is received by MATLAB as the input to the algorithm.
2. MATLAB identifies consecutive periods by autocorrelating over a small-time window and noting the lag associated with the maximum autocorrelation.

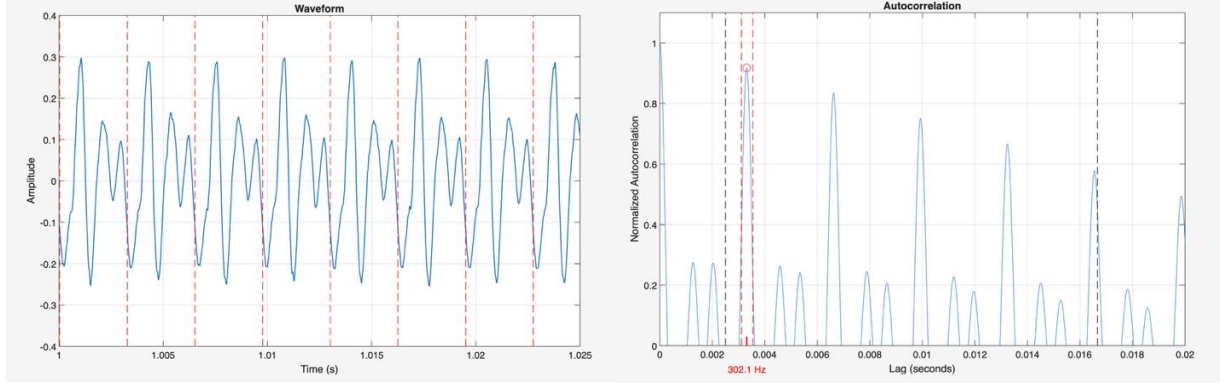


Figure 2: The waveform plot (left) marks each consecutive period found in the audio file. The autocorrelation plot (right) shows how each period is identified.

3. From the newly produced array of consecutive periods, the peak-to-peak amplitude in each period is saved in a separate array. The array of consecutive periods and consecutive peak-to-peak amplitudes are used to compute local jitter and shimmer, respectively.

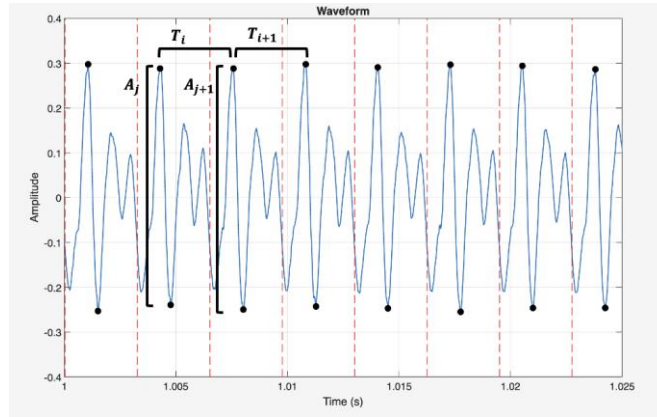


Figure 3: The waveform plot (left) identifies the difference in consecutive periods and consecutive periods peak-to-peak amplitudes that local jitter and local shimmer measure, respectively.

Frequency Domain Analysis

4. The fundamental frequency is computed from the consecutive periods array and the locations of the audio's harmonics are determined by detecting peaks around harmonic estimations. Harmonic bandwidth was determined by computing the Power Spectral Density (PSD) and identifying the full-width-half-max bounds of the harmonic for each harmonic peak of the PSD. Using the tuned

harmonic bandwidth and the audio's power spectral density, the harmonic power and noise power are used to compute HNR and per second power contribution.

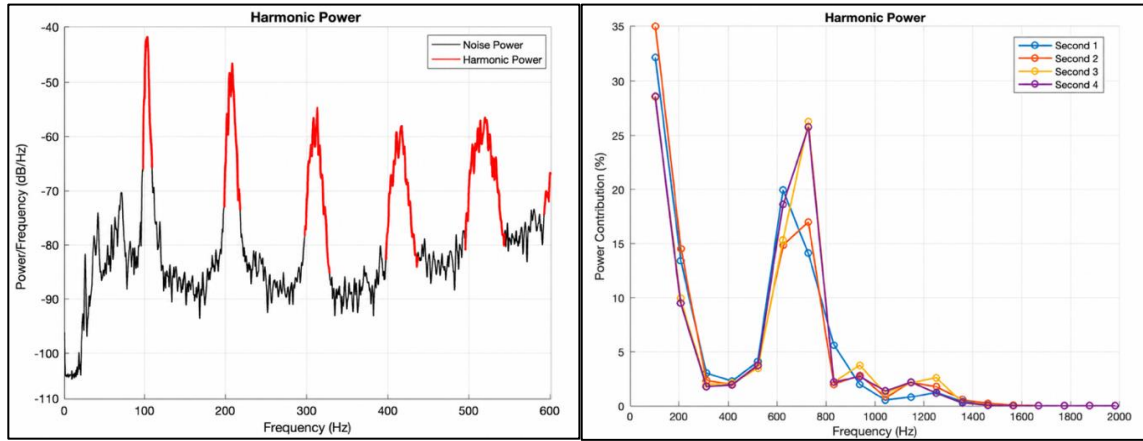


Figure 4: The harmonic power plot (left) highlights the harmonic power (red) and noise power (black) used in the HNR metric. The per second harmonic power contribution plot (right) shows the harmonic power distribution for one second intervals across a four second audio file.

IV. Results

The following results were obtained using metrics generated by our algorithm on a subset of 46 files from the ALS and non-ALS databases, each with a sampling rate of 44,100 Hz and a recording duration of at least 3 seconds.

1. Perturbation and HNR of ALS vs Non-ALS prolonged vowels

ALS patients had higher values of all the perturbation metrics and lower values for the static metric. These results reflect physiological expectations; lower neuromuscular control results in more perturbation and lower HNR.

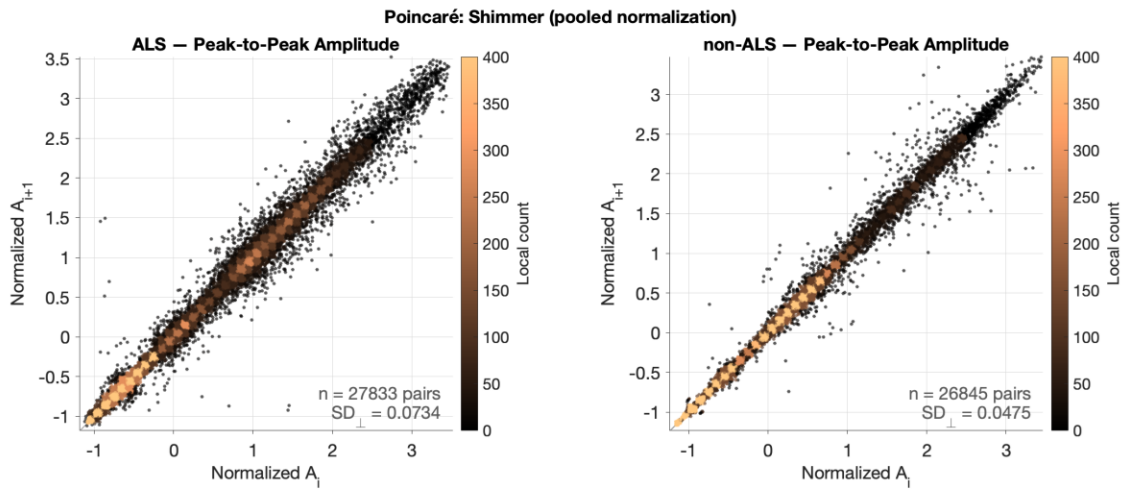


Figure 5: Poincaré plot of normalized z-score peak-to-peak amplitude of next period vs current period. ALS shown on right; Non-ALS on left.

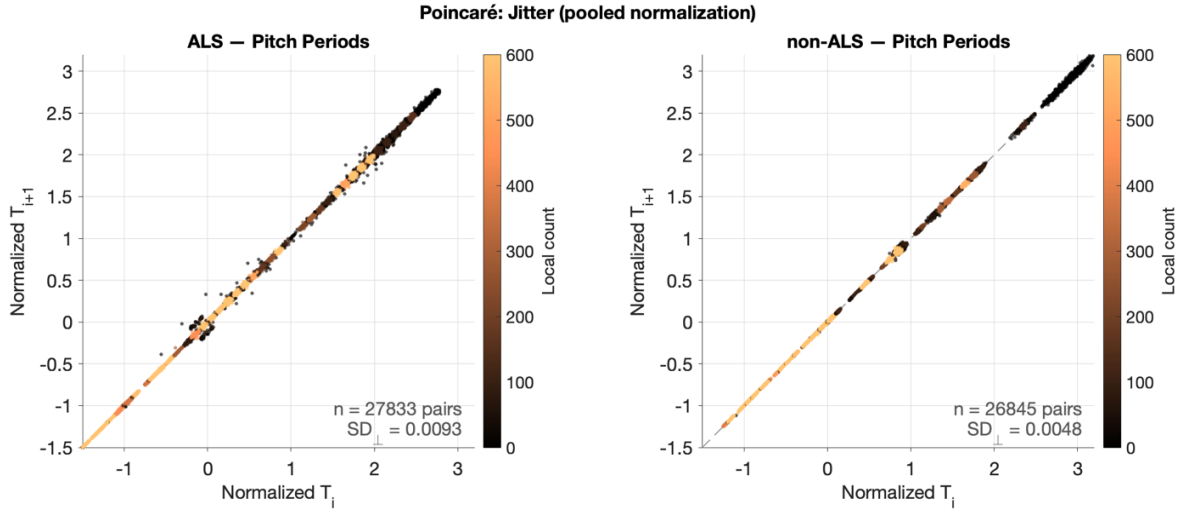


Figure 6: Poincaré plot of normalized z-score next period length plotted against current period length. ALS shown on right; Non-ALS on left.

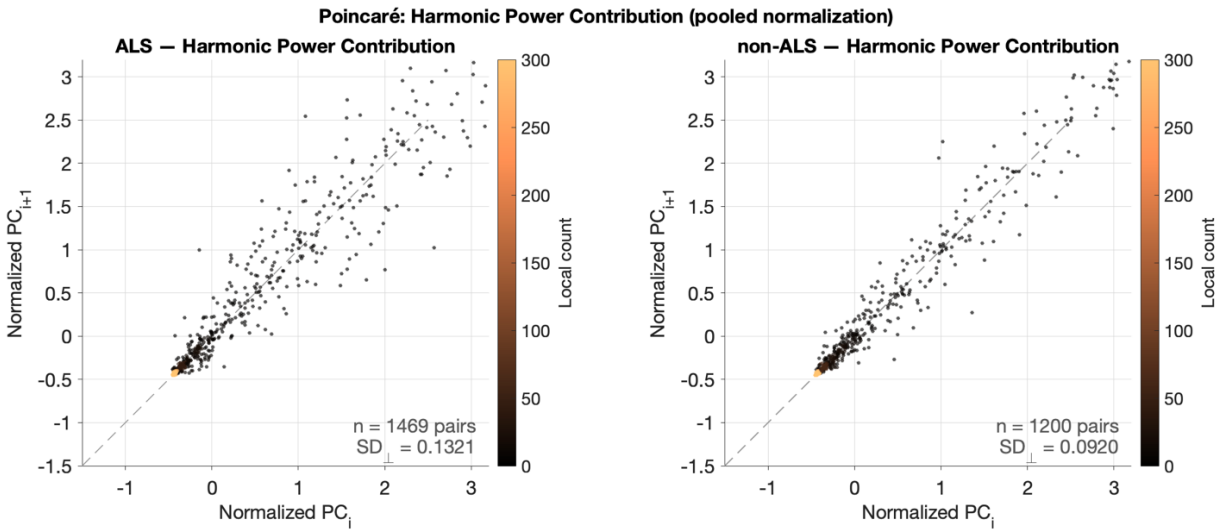


Figure 7: Poincaré plot of normalized z-score power contribution of next second vs current second across harmonics below 2kHz. ALS shown on right; Non-ALS on left.

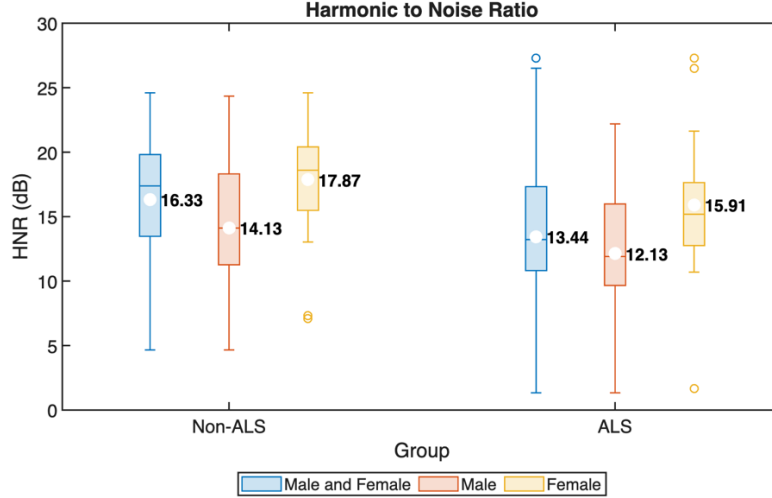


Figure 8: Box plots showing HNR population statistics for non-ALS and ALS groups, including sex-specific comparisons. The mean HNR value for each group is shown as a dot with its corresponding numerical value.

2. Characterizing ALS and Non-ALS individuals by metrics

Using a composite metric approach, our system differentiated ALS from non-ALS individuals with an AUC of 0.733. Characterization performance varied by sex, with an AUC of 0.74 for female participants and 0.67 for male participants.

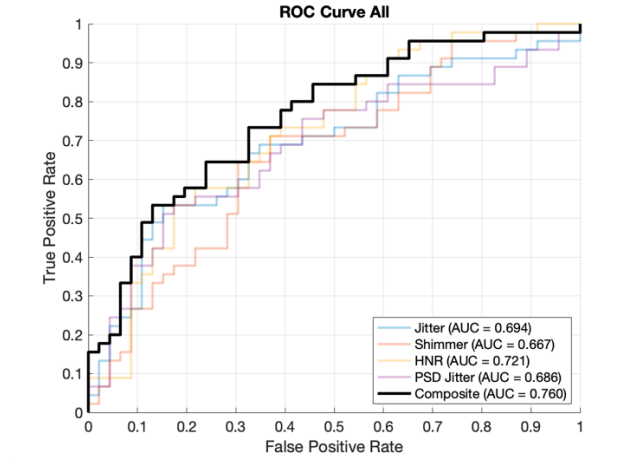


Figure 9: ROC describing characterization of ALS individuals from 46 ALS and 46 Non-ALS prolonged vowel files from the Minsk2020 ALS Database. Area under the curve (AUC) is shown for all individual metrics, as well as for a combined metric generated by weighting each metric according to its AUC individual performance. False positives were defined as cases in which an individual was incorrectly identified as having ALS when they were non-ALS.

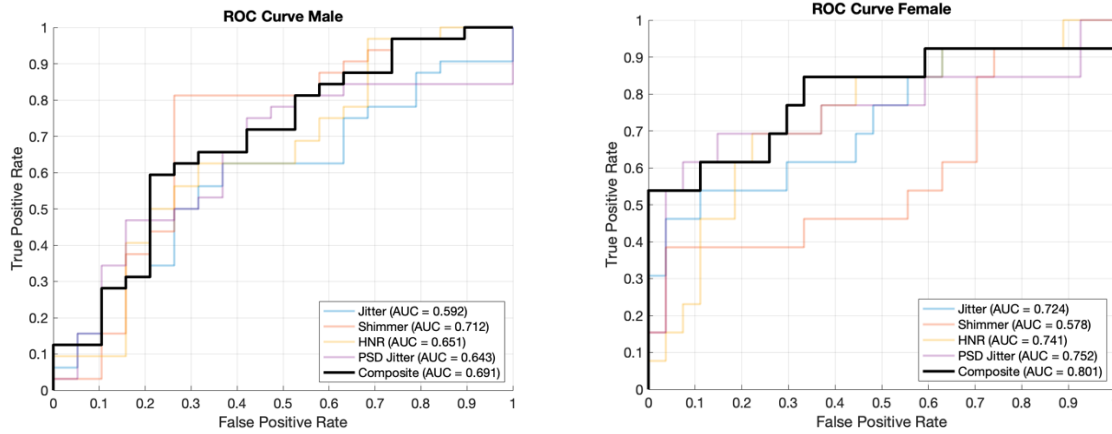


Figure 10: ROC curves characterizing ALS versus non-ALS prolonged vowel recordings from the Minsk2020 ALS Database, shown separately for female participants (14 ALS, 27 non-ALS; right) and male participants (33 ALS, 19 non-ALS; left). AUC is reported for each individual metric and for a composite metric generated by weighting each metric according to its individual AUC performance. False positives were defined as non-ALS individuals incorrectly classified as having ALS.

V. Discussion

Running our algorithm on the Minsk2020 ALS Database produced several meaningful results. First, perturbation metrics, including jitter, shimmer, and per-second harmonic power contribution, were higher in the ALS group compared with the non-ALS group, as shown in Figures 5–7. Each figure is shown as a Poincaré plot, where tighter clustering around the centerline indicates lower perturbation, while greater spread indicates increased perturbation and reduced vocal consistency. This increased inconsistency across the prolonged vowel recordings is also reflected by the standard deviation of each Poincaré plot, calculated as the average perpendicular distance of each point from the centerline. For jitter, shimmer, and per-second harmonic power contribution, the group-wise standard deviation was larger for ALS than for non-ALS. This pattern is consistent with prior literature using this database, as well as the theoretical expectation that neuromuscular degeneration in ALS reduces an individual’s ability to maintain consistent phonation during prolonged vowels [5].

Harmonic-to-noise ratio was higher in the non-ALS group than in the ALS group for both male and female participants, as shown in Figure 48. This result is also consistent with expectations, as ALS-related degeneration of the vocal system can increase noise during phonation due to inadequate vocal fold closure and aperiodic vocal fold vibration, thereby reducing the harmonic-to-noise ratio [8]. Together, these results demonstrate the ability of our algorithm to quantify physiological control of the vocal system using jitter, shimmer, harmonic-to-noise ratio, and per-second harmonic power contribution.

Also, receiver operating characteristic (ROC) curves further demonstrate the algorithm’s ability to characterize ALS-related degeneration in the vocal system. The area under the curve (AUC) quantifies each metric’s ability to distinguish ALS from non-ALS recordings. Using a composite metric across the full ALS and non-ALS population, the algorithm achieved an AUC of 0.76, as shown in Figure 9. Sex-specific results also showed meaningful characterization performance, with an AUC of 0.69 for male participants (Figure 10: left) and 0.80 for female participants (Figure 10: right). Overall, these results

suggest that the extracted vocal biomarkers are useful for measuring ALS-related degeneration in the vocal system.

A major limitation of the system is its reliance on the assumption that vocal signals are quasi-periodic. This assumption may break down for highly degenerated voices, making the algorithm less robust for severe cases. Because the method quantifies inconsistency relative to a periodic baseline, errors can arise in consecutive period detection and harmonic bandwidth estimation. These errors may reduce the reliability of jitter, shimmer, HNR, and per-second harmonic power contribution measurements.

VI. Conclusions

Overall, the program robustly computes vocal metrics including jitter, shimmer, HNR, and per-second harmonic power contribution. These metrics computed were shown to be meaningful ways to quantify differences in the vocal systems between ALS and Non-ALS individuals. The code for these functions is publicly available and is intended to support future research on vocal biomarkers of disease and potential clinical applications.

VII. Deliverables

Table I: Proposed deliverables and associated status.

Proposed Deliverable	Completed
MATLAB algorithm that accepts inputs and outputs vocal biomarkers	✓
Interactive demo of vocal biomarkers	✓
Final report	✓
Webpage for project	✓

VIII. Timeline

Project Development Schedule												
Week:	1	2	3	4	5	6	7	8	9	10	11	12
	Build REV1 of algorithm for non-ALS: Jitter, Shimmer, HNR, Fundamental period, <i>Formants</i>											
				- Per-second harmonic contribution - Revise algorithm, produce REV2 for non-ALS: Jitter, Shimmer, HNR, <i>Formants</i>								
									ALS/non-ALS characterization			
										Prepare demo and poster		
										Prepare final paper and webpage		
Deliverables:	PR 4	PR 5 Proposal revision	PR 6	PR 7 Video	PR 8	PR 9	PR 10	PR 11	PR 12	PR 13	DEMO	Paper Webpage

Figure 11: Proposed Gantt chart annotated to reflect changes throughout the development process. Red strike-through items indicate steps removed from the workflow, while red items without strike-through denote additions to the process.

Appendix A

a. Relating to prior course work

This capstone can be seen as a direct extension of the work completed in ESE 4880 Signals & Imaging Laboratory Module 1. The project incorporates a similar microphone/soundcard setup for a live-demo and MATLAB for algorithm development, focusing on expanded feature extraction instead of purely cross-correlation. Very similarly, ESE 4820 Digital Signal Processing the principle of digital signal processing such as the FFT algorithm, up sampling, and other signal processing techniques. This capstone uses the FFT algorithm to characterize frequency components of the audio signals and up sampling the audio to increase measuring sensitivity. Finally, ESE 351 Signals and Systems taught many fundamental signal processing methods are derived from the basic core principle taught in ESE 351.

b. Relevant Engineering Standards

The coding platform used in this capstone is MATLAB used for processing audio and extracting features. Additionally, the capstone follows standard biomedical signal processing practices such as preprocessing, feature extraction, and statistical comparison between subject groups.

In addition to technical standards, the capstone adheres to ABET student objectives. The capstone project applies engineering principles to solve a real-world biomedical signal processing problem. The design and implementation of the algorithm is under practical constraints such as noise and limited clinical access. Experimental analysis is performed through feature extraction and statistical comparison between individuals with ALS and no ALS to evaluate system performance. Additionally, ethical responsibility and clear technical communication are emphasized due to the medical context and the possibility to advance ALS research.

c. Constraints

The dataset used in the project is open source due to the legal challenges of accessing raw audio files from many ALS databases. Although the open-source dataset helps simulate imperfect audio files for realistic applications, there is no guarantee that this dataset is high quality. A subset of the audio files is used, removing distorted audio files.

Appendix B

Project team: The project team is composed of Washington University undergraduates Yvette Gerber and Hayden Somach. Yvette earned a BA in Science, Technology, and Society from the University of Redlands and is now pursuing a BS in Systems Science & Engineering at Washington University in St. Louis. Her work focuses on signal processing and medical imaging, supported by her coursework and laboratory experience at WashU. Hayden received a BS in Physics from Grinnell College and is currently pursuing a degree in Electrical Engineering at WashU, with a focus on radio-frequency systems and signal processing. They share a strong background in signal processing and previously collaborated on a semester-long Signals and Imaging laboratory course, where they developed an algorithm foundational to

the proposed project. Given their overlapping strengths and interests, the team will collaborate throughout the full scope of the project.

The project took place under ESE 4970 Electrical Engineering Capstone Design Projects with advisors Jason Trobaugh and Matthew Lew. Professor Trobaugh and Professor Lew were invested in the project demonstrating the ability of Yvette and Hayden in the context of degree requirements as well as benefiting the Department of Electrical and Systems Engineering. The project client is Zachary Abrams, who has significant experience with vocal biomarkers within the B2AI project. Zachary Abrams was invested in the contribution of this project to advance the field of voice as a biomarker, especially for neurodegenerative conditions such as ALS. Through a better understanding of how voice as biomarkers can be measured and detected in ALS, he hopes to better identify subgroups, treatment set-ups, and earlier detection of the disease over time. Throughout the duration ESE 4970 Electrical Engineering Capstone Design course, there were weekly 30-minute meetings with the advisors, and biweekly hour-long meetings with our client.

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